Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
MTH174	ENGINEERING MATHEMATICS	3	1	0	4	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: recall the concept of matrices and their applications to solve the system of linear equations.

CO2 :: understand the use of different methods for the solution of linear differential equations.

CO3 :: understand the elementary notions of Fourier series for harmonic analysis.

CO4 :: apply the concept of multi-variable differential calculus for solving problems in the field of sciences and engineering.

CO5 :: analyze the surface and volume integrals using various concepts of multi-variable integral calculus.

	TextBooks (T)		
Sr No	**Title**	**Author**	**Publisher Name**
T-1	ADVANCED ENGINEERING MATHEMATICS	R.K.JAIN, S.R.K. IYENGER	NAROSA PUBLISHING HOUSE
	Reference Books (R)		
Sr No	**Title**	**Author**	**Publisher Name**
R-1	HIGHER ENGINEERING MATHEMATICS	B.S. GREWAL	KHANNA PUBLISHERS

Other Reading (OR)	
Sr No	**Journals articles as Compulsary reading (specific articles, complete reference)**
OR-1	https://ncert.nic.in/textbook.php?lemh1=3-6 ,

Relevant Websites (RW)		
Sr No	**(Web address) (only if relevant to the course)**	**Salient Features**
RW-1	https://tutorial.math.lamar.edu/Classes/CalcIII/PartialDerivatives.aspx	Lecture notes on Partial Derivatives
RW-2	https://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx	Lecture notes on multiple integral
RW-3	https://tutorial.math.lamar.edu/Classes/DE/IntroSecondOrder.aspx	Second order linear differential equations
RW-4	https://tutorial.math.lamar.edu/Classes/DE/HOHomogeneousDE.aspx	Homogeneous linear differential equations with constant coefficients

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RW-5	http://tutorial.math.lamar.edu/Classes/DE/NonhomogeneousDE.aspx	Non homogeneous differential equations
RW-6	http://tutorial.math.lamar.edu/Classes/DE/UndeterminedCoefficients.aspx	Method of undetermined coefficients
RW-7	http://tutorial.math.lamar.edu/Classes/DE/VariationofParameters.aspx	Method of variation of parameters
RW-8	https://mathworld.wolfram.com/FourierSeries.html	Fourier series

Audio Visual Aids (AV)		
Sr No	**(AV aids) (only if relevant to the course)**	**Salient Features**
AV-1	https://www.khanacademy.org/math/linear-algebra/	Video lecture on inverse of matrices and solution of system of linear equations.
AV-2	https://www.khanacademy.org/math/multivariable-calculus	Video lectures on multivariable-calculus
AV-3	https://www.youtube.com/watch?v=4rc3w1sGoNU&list=UUMMt9zLt3UojrK2z52E5vng&index=2	NPTEL Video lectures on Change of Order of Integration by IIT Roorkee
AV-4	https://www.youtube.com/watch?v=vA9dfINW4Rg	video lectures on Fourier series
AV-5	https://www.youtube.com/watch?v=GoyeNUaSW08	Lagrange multipliers
AV-6	https://www.youtube.com/watch?v=RglnnEVaVTI	Video lecture on Double Integrals in Polar Form
AV-7	https://youtu.be/MEsOhv1Wubc	Cayley Hamilton Theorem
AV-8	https://www.youtube.com/watch?v=3kprJPsgGpM	Solution of Higher Order Homogeneous Linear Equations
AV-9	https://archive.nptel.ac.in/courses/111/105/111105134/	Fourier series for Even and Odd function.
AV-10	https://archive.nptel.ac.in/courses/111/105/111105121/	Derivative of Composite function.

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	**Lecture Number**	**Broad Topic(Sub Topic)**	**Chapters/Sections of Text/reference books**	**Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs**	**Lecture Description**	**Learning Outcomes**	**Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned**	**Live Examples**

Week 1	Lecture 1	Matrix Algebra(elementary operations and their use in getting the rank)	T-1 R-1	OR-1	Lecture Zero and Review of Matrix in Lecture 1 and Rank of matrix in Lecture 2	Recall the basic concepts of matrix and explain rank of matrix using elementary operations.	Discussion and Brainstorming .	Situational Problem: 1.Consider each study chair of your class room as an element of a matrix. Find the order of the matrix formed by your class room. What are the other possible orders? 2. Try to find out various arrangements of things around you that can be presented as a matrix. 3. Which special kind of matrix arrangement is being followed by any wing of army while doing parade.

Week 1	Lecture 2	Matrix Algebra(elementary operations and their use in getting the rank)	T-1 R-1	OR-1	Lecture Zero and Review of Matrix in Lecture 1 and Rank of matrix in Lecture 2	Recall the basic concepts of matrix and explain rank of matrix using elementary operations.	Discussion and Brainstorming .	Situational Problem: 1.Consider each study chair of your class room as an element of a matrix. Find the order of the matrix formed by your class room. What are the other possible orders? 2. Try to find out various arrangements of things around you that can be presented as a matrix. 3. Which special kind of matrix arrangement is being followed by any wing of army while doing parade.
	Lecture 3	Matrix Algebra(inverse of a matrix and solution of linear simultaneous equations)	T-1 R-1	AV-1	In Lecture 3, Inverse of matrix will be discussed. In Lecture 4, solution of linear simultaneous equations will be discussed.	Learn the conditions for the existence of a unique solution, no solution, and infinitely many solutions in homogeneous and non-homogeneous linear simultaneous equations and understand the method to find the inverse of a matrix.	Problem solving and Brainstorming	Situational Problem: Vani, her father and her grandfather have an average age of 53. One-half of her grandfather's age plus one-third of her father's age plus one fourth of Vani's age is 65. Four years ago, if Vani's grandfather was four times as old as Vani, then how old are they all now?

Week 2	Lecture 4	Matrix Algebra(inverse of a matrix and solution of linear simultaneous equations)	T-1 R-1	AV-1	In Lecture 3, Inverse of matrix will be discussed. In Lecture 4, solution of linear simultaneous equations will be discussed.	Learn the conditions for the existence of a unique solution, no solution, and infinitely many solutions in homogeneous and non-homogeneous linear simultaneous equations and understand the method to find the inverse of a matrix.	Problem solving and Brainstorming	Situational Problem: Vani, her father and her grandfather have an average age of 53. One-half of her grandfather's age plus one-third of her father's age plus one fourth of Vani's age is 65. Four years ago, if Vani's grandfather was four times as old as Vani, then how old are they all now?

Week 2	Lecture 5	Matrix Algebra(eigen-values and eigenvectors of a matrix)	T-1 R-1		Calculation of eigenvalues and eigenvectors from characteristic equation of given matrix.	Discuss the method to find the eigenvalues and eigenvectors of a given system under the given matrix transformation and apply the properties of Eigen values in various scientific fields.	Discussion and Brainstorming	Situational Problem: 1. How the melting of butter on bread is related with the concept of eigen values and eigen vectors? 2. The eigenvalues can also be used to determine if a structure has deformed under the application of a particular force. eigenvalues for the structure are measured before and after the application of force. If a change in the eigenvalues is observed, it means the structure has undergone deformation.

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Week 2	Lecture 6	Matrix Algebra(eigen-values and eigenvectors of a matrix)	T-1 R-1		Calculation of eigenvalues and eigenvectors from characteristic equation of given matrix.	Discuss the method to find the eigenvalues and eigenvectors of a given system under the given matrix transformation and apply the properties of Eigen values in various scientific fields.	Discussion and Brainstorming	Situational Problem: 1. How the melting of butter on bread is related with the concept of eigen values and eigen vectors? 2. The eigenvalues can also be used to determine if a structure has deformed under the application of a particular force. eigenvalues for the structure are measured before and after the application of force. If a change in the eigenvalues is observed, it means the structure has undergone deformation.
Week 3	Lecture 7	Matrix Algebra(Cayley-Hamilton theorem)	T-1 R-1	AV-7	Cayley Hamilton theorem	Use Cayley Hamilton theorem in finding the inverse of matrix in an easy way.	Teaching and learning through Flipped Classroom	Cayley Hamilton theorem is applicable to which of the following matrices: (a) Row matrix (b) Column matrix (c) Rectangular matrix (d) Square

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Week 3	Lecture 8	Linear differential equation-I(introduction to linear differential equation)	T-1	RW-3	Classification of linear differential equations as homogeneous or non-homogeneous and constant coefficient or variable coefficients and Condition of existence and uniqueness of solution of IVP.	Learn about the classification of linear differential equations and understand the concept of the finding the intervals for the differential equation to be normal.	Lecture cum discussion	Second order linear mathematical model can be solved for the solution of problem where a mass weighing 3 lbs. stretch a spring (which is 4 ft. long) 3 inches. If the mass is raised 1 inch above its equilibrium position and given an initial velocity of 2 ft./sec. upward, determine the subsequent motion (i.e. find the distance from the equilibrium position as a function of time). Assume that the air resistance is negligible.

Week 3	Lecture 8	Linear differential equation-I(solution of linear differential equation)	T-1	RW-3	Classification of linear differential equations as homogeneous or non-homogeneous and constant coefficient or variable coefficients and Condition of existence and uniqueness of solution of IVP.	Learn about the classification of linear differential equations and understand the concept of the finding the intervals for the differential equation to be normal.	Lecture cum discussion	Second order linear mathematical model can be solved for the solution of problem where a mass weighing 3 lbs. stretch a spring (which is 4 ft. long) 3 inches. If the mass is raised 1 inch above its equilibrium position and given an initial velocity of 2 ft./sec. upward, determine the subsequent motion (i.e. find the distance from the equilibrium position as a function of time). Assume that the air resistance is negligible.
	Lecture 9				Test 1			

Week 4	Lecture 10	Linear differential equation-I(linear dependence and linear independence of solution)	T-1	AV-5	linear dependence and linear independence of solution	Describe the term Wronskian to check the linear dependence and independence of functions.	Lecture cum discussion.	Intuitively vectors being linearly independent means they represent independent directions in your vector spaces, while linearly dependent vectors don't. For example, if you have a set of vectors {x1, x2, x3} and you can walk distance a1 in the x1 direction, then some distance a2 in x2, then again a3 in the direction of x3. If in the end you are back where you started, then the vectors are linearly dependent.
	Lecture 11	Linear differential equation-I(method of solution of linear differential equation-differential operator)	T-1 R-1		Differential operator D=d/dx and the Symbolic form of differential equations.	Learn about the use of differential operator in writing the symbolic form of differential equations.	Lecture cum discussion.	
	Lecture 12	Linear differential equation-I(solution of second order homogeneous linear differential equation with constant coefficient)	T-1 R-1	RW-3	Auxiliary equation using differential operator and General solution of second order linear homogeneous differential equation with constant coefficients.	Remember the rules of writing the general solution of differential equation when the roots of quadratic auxiliary equation are real (distinct or equal) and complex.	Lecture cum discussion.	

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Week 5	Lecture 13	Linear differential equation-I(solution of higher order homogeneous linear differential equations with constant coefficient)	T-1 R-1	RW-4 AV-8	General solution of higher order linear homogeneous differential equation with constant coefficients.	Remember the rules of writing the general solution of differential equation when the roots of cubic, fourth degree auxiliary equation are multiple real or multiple complex.	Teaching and learning through Flipped Classroom	
	Lecture 14	Linear differential equation-II(solution of non-homogeneous linear differential equations with constant coefficients using operator method)	T-1 R-1	RW-5	Particular integral for non-homogeneous differential equations using operator method.	Learn the method to find particular integral if the right-side function is of the form X=e^ax and X=sin(ax+b) or cos (ax+b) along with case of failure using operator method.	Lecture cum discussion.	
	Lecture 15	Linear differential equation-II(solution of non-homogeneous linear differential equations with constant coefficients using operator method)	T-1 R-1	RW-5	Particular integral for non-homogeneous differential equations using operator method.	Learn the method to find particular integral if the right-side function is of the form X=e^ax and X=sin(ax+b) or cos (ax+b) along with case of failure using operator method.	Lecture cum discussion.	
Week 6	Lecture 16	Linear differential equation-II(method of variation of parameters)	T-1 R-1	RW-7	Particular integral for non-homogeneous differential equations using variation of parameter.	Understand the method to find particular integral using variation of parameter.	Lecture cum discussion.	
	Lecture 17				Test 2			
	Lecture 18	Linear differential equation-II(method of undetermined coefficient)	T-1 R-1	RW-6	Particular integral for non-homogeneous differential equations using method of undetermined coefficients.	Learn to choose correct form of particular integral depending upon right side function and complementary function in method of undetermined coefficients.	Lecture cum discussion.	

Week 7	Lecture 19	Linear differential equation-II(solution of Euler-Cauchy equation)	T-1 R-1		Euler-Cauchy differential equation	Know the rules to convert this special type of differential equation with variable coefficients to an equation with constant coefficients and apply the previous known methods to find general solutions.	Lecture cum discussion.	
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Fourier Series(introduction and Euler's formulae)	T-1 R-1	RW-8 AV-4	Fourier series and Euler's formula	Remember the Euler formula to find Fourier coefficients and hence Fourier series	Discussion and Brainstorming.	
	Lecture 23	Fourier Series(conditions for a Fourier expansion and functions having points of discontinuity)		RW-8 AV-4	Conditions for a Fourier Expansion and Functions having points of discontinuity	Understand the conditions for a Fourier expansion and learn to write the Fourier series for functions having points of discontinuity	Problem solving and Brainstorming.	
	Lecture 24	Fourier Series(change of interval)	T-1 R-1	RW-8 AV-4	Change of interval	Learn the method to find the Fourier. series for Change of interval	Discussion and Brainstorming	
Week 9	Lecture 25	Fourier Series(even and odd functions)	T-1 R-1	AV-9	Fourier series for even and odd functions	Understand the terms even and odd function and learn to write the Fourier series for these.	Teaching and learning through Flipped Classroom	
	Lecture 26	Fourier Series(half range series)	T-1 R-1		Fourier half range series.	Learn methods to find the Fourier series for half range.	Discussion and Brainstorming	

Week 9	Lecture 27	Multivariate Calculus(limit)	T-1	RW-1 AV-2	Functions of two variables, Limit, Continuity and Total Derivative	Learn the extension of the concept of limit, continuity and differentiability of single variable to multiple variables.	Discussion and Brainstorming.	
		Multivariate Calculus (continuity and differentiability of functions of two variables)	T-1	RW-1 AV-2	Functions of two variables, Limit, Continuity and Total Derivative	Learn the extension of the concept of limit, continuity and differentiability of single variable to multiple variables.	Discussion and Brainstorming.	
Week 10	Lecture 28	Multivariate Calculus(limit)	T-1	RW-1 AV-2	Functions of two variables, Limit, Continuity and Total Derivative	Learn the extension of the concept of limit, continuity and differentiability of single variable to multiple variables.	Discussion and Brainstorming.	
		Multivariate Calculus (continuity and differentiability of functions of two variables)	T-1	RW-1 AV-2	Functions of two variables, Limit, Continuity and Total Derivative	Learn the extension of the concept of limit, continuity and differentiability of single variable to multiple variables.	Discussion and Brainstorming.	
	Lecture 29	Multivariate Calculus(chain rule)	T-1 R-1	AV-10	Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Teaching and learning through Flipped Classroom	
		Multivariate Calculus (change of variables)	T-1 R-1		Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Discussion and Brainstorming	
		Multivariate Calculus (Jacobians)	T-1 R-1		Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Discussion and Brainstorming	

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Week 10	Lecture 30	Multivariate Calculus(chain rule)	T-1 R-1	AV-10	Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Teaching and learning through Flipped Classroom	
		Multivariate Calculus (change of variables)	T-1 R-1		Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Discussion and Brainstorming	
		Multivariate Calculus (Jacobians)	T-1 R-1		Chain rule of differentiation and change of variables using Jacobian.	Use of chain rule to evaluate the derivative of composite function and learn the concept of Jacobians in change of variables.	Discussion and Brainstorming	
Week 11	Lecture 31	Multivariate Calculus (Euler’s theorem for homogeneous equations)	T-1 R-1		Euler's theorem for Homogeneous functions	Understand the term Homogeneous functions and discuss Euler's theorem on homogeneous functions.	Discussion and Brainstorming	
	Lecture 32	Multivariate Calculus (extrema of functions of two variables)	T-1		extrema of functions of two variables	Discuss the rules to find points of extrema in function of two variables	Discussion and Brainstorming.	
	Lecture 33	Multivariate Calculus (Lagrange’s method of undetermined multipliers)	T-1	AV-5	Lagrange’s method of undetermined multipliers	Explain optimization of function under the given constraints.	Problem solving	
Week 12	Lecture 34				Test 3			
	Lecture 35	Integral Calculus(double integrals)	T-1 R-1		Double integrals	Learn the extension of idea of single integral to double integral over two-dimensional region.	Discussion and Brainstorming.	
	Lecture 36	Integral Calculus(change of order of integration)	T-1 R-1	RW-2 AV-3	Change of order of integration	Discuss about no change in value of integral on changing order of integration.	Discussion and Brainstorming.	

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Week 13	Lecture 37	Integral Calculus(application of double integrals to calculate area and volume)	T-1 R-1	RW-2	Application of double integrals to calculate area and volume	Apply double integral to find area and volume	Problem solving	
	Lecture 38	Integral Calculus(triple integrals)	T-1 R-1	RW-2	Triple integrals.	Learn the evaluation of triple integral over three-dimensional space.	Discussion and Brainstorming.	
	Lecture 39	Integral Calculus(change of variables)	T-1 R-1	RW-2 AV-6	Change of variables and Application of triple integral to calculate volume.	Learn the concept of change of variables and apply triple integral to calculate volume.	Discussion and Brainstorming.	
		Integral Calculus(application of triple integrals to calculate volume)	T-1 R-1	RW-2 AV-6	Change of variables and Application of triple integral to calculate volume.	Learn the concept of change of variables and apply triple integral to calculate volume.	Discussion and Brainstorming.	
Week 14	Lecture 40	Integral Calculus(change of variables)	T-1 R-1	RW-2 AV-6	Change of variables and Application of triple integral to calculate volume.	Learn the concept of change of variables and apply triple integral to calculate volume.	Discussion and Brainstorming.	
		Integral Calculus(application of triple integrals to calculate volume)	T-1 R-1	RW-2 AV-6	Change of variables and Application of triple integral to calculate volume.	Learn the concept of change of variables and apply triple integral to calculate volume.	Discussion and Brainstorming.	
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test 2	50	CO2
Test 1	50	CO1

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Test 3	50	CO3, CO4

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test 1	To evaluate the conceptual clarity of the students about Matrix Algebra using subjective test of 30 marks.	elementary operations and their use in getting the rank, inverse of a matrix and solution of linear simultaneous equations, eigen-values and eigenvectors of a matrix, Cayley-Hamilton theorem.	Individual	Offline	30	2 / 3
Test 2	To evaluate the conceptual clarity of the students about Linear Differential Equations using subjective test of 30 marks.	introduction to linear differential equation, solution of linear differential equation, linear dependence and linear independence of solution, method of solution of linear differential equationdifferential operator, solution of second order homogeneous linear differential equation with constant coefficient, solution of higher order homogeneous linear differential equations with constant coefficients, solution of non-homogeneous linear differential equations with constant coefficients using operator method.	Individual	Offline	30	5 / 6
Test 3	To evaluate the conceptual clarity of the students about Unit IV and V using subjective test of 30 marks.	introduction and Euler's formulae, conditions for a Fourier expansion and functions having points of discontinuity, change of interval, even and odd functions, half range series, limit, continuity and differentiability of functions of two variables, chain rule, change of variables, Euler’s theorem for homogeneous equations, Jacobians	Individual	Offline	30	11 / 12

MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
Test 1	ENGINEERING MATHEMATICS - I	MOOCs	NPTEL
Test 2	ENGINEERING MATHEMATICS - I	MOOCs	NPTEL
Test 3	ENGINEERING MATHEMATICS - I	MOOCs	NPTEL

Where MOOCs/ Certification etc. are mapped with Academic Tasks:
1. Students have choice to appear for Academic Task or MOOCs etc.
2. The student may appear for both, In this case best obtained marks will be considered.

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	elementary operations and their use in getting the rank, inverse of a matrix and solution of linear simultaneous equations.	Problem solving and discussion on MCQ questions
Tutorial2	eigen-values and eigenvectors of a matrix, Cayley-Hamilton theorem	Problem solving and discussion on MCQ questions
Tutorial3	Solution of linear differential equation, linear dependence and linear independence of solution, method of solution of linear differential equation- differential operator	Problem solving and discussion on MCQ questions
Tutorial4	Solution of second order homogeneous linear differential equation with constant coefficient, solution of higher order homogeneous linear differential equations with constant coefficient	Problem solving and discussion on MCQ questions
Tutorial5	Solution of non-homogeneous linear differential equations with constant coefficients using operator method	Problem solving and discussion on MCQ questions
Tutorial6	Method of variation of parameters, method of undetermined coefficient	Problem solving and discussion on MCQ questions
Tutorial7	Solution of Euler-Cauchy equation	Problem solving and discussion on MCQ questions
After Mid-Term		
Tutorial8	Euler's formulae, Conditions for a Fourier Expansion and Functions having points of discontinuity, Change of interval.	Problem solving and discussion on MCQ questions
Tutorial9	Even and odd functions, Half Range series	Problem solving and discussion on MCQ questions
Tutorial10	Limit and Continuity, Partial derivatives, Total derivative and differentiability	Problem solving and discussion on MCQ questions
Tutorial11	Chain rule, Euler's theorem for Homogeneous functions, Maxima and Minima, Lagrange method of multiplier.	Problem solving and discussion on MCQ questions
Tutorial12	Double integrals, change of order of integration	Problem solving and discussion on MCQ questions
Tutorial13	Triple integral, Change of variable	Problem solving and discussion on MCQ questions
Tutorial14	Application of double integrals to calculate area and volume, Application of triple integrals to calculate volume.	Problem solving and discussion on MCQ questions

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